

Supplementary Winter Examination – 2024

Course: B.Tech Branch : Electrical Engineering & Allied Semester : IV

Subject Code & Name: (BTEEC401) & Network Theory

Max Marks: 60

Date: 19/12/2024

Duration: 3 Hr.

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per COBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

(Level/ CO) Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- 1 Four resistances $80\ \Omega$, $50\ \Omega$, $25\ \Omega$, and R are connected in parallel. Current through $25\ \Omega$ resistor is 4 A. Total current of the supply is 10 A. The value of R will be

CO1 1

- (a) $66.66\ \Omega$ (b) $40.25\ \Omega$ (c) $76.56\ \Omega$ (d) $36.36\ \Omega$

- 2 Two wires A and B of the same material and length L and $2L$ have radius r and $2r$ respectively. The ratio of their specific resistance will be

CO2 1

- (a) 1 : 1 (b) 1 : 2 (c) 1 : 4 (d) 1 : 8

- 3 A network contains only an independent current source and resistors. If the values of all resistors are doubled, the value of the node voltages will

CO3 1

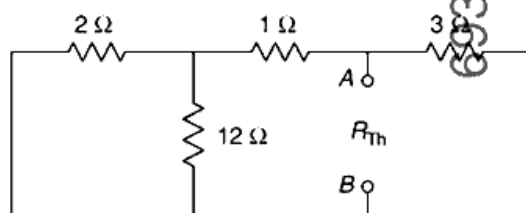
- (a) become half (b) become double (c) remain unchanged (d) none of these

- 4 Superposition theorem is not applicable to networks containing

CO3 1

- (a) nonlinear elements (b) dependent voltage source (c) dependent current source (d) Transformers

- 5 Determine Thevenin's equivalent resistance:



CO4 1

- (a) 1.46 ohm (b) 1.43 ohm (c) 1.5 ohm (d) 3 ohm

- 6 In a series RLC high Q circuit, the current peaks at a frequency

CO4 1

- (a) equal to the resonant frequency (b) greater than the resonant frequency (c) less than the resonant frequency (d) none of the above.

7 As the poles of a network shift away from the x axis, the response

- (a) remains constant (b) becomes less oscillating (c) becomes more oscillating (d) none of these

CO5 1

8 Which of the following describes a time-invariant system?

- a) The system parameters do not change with time.
b) The system parameters vary periodically with time.
c) The system is linear and bilateral.
d) The system is reciprocal.

CO4 1

9 For a two-port network to be reciprocal

- (a) $Z_{11} = Z_{22}$ (c) $AD - BC = 1$
(b) $y_{12} = y_{21}$ (d) $h_{12} = -h_{21}$

CO3 1

10 Two two-port networks are connected in cascade. The combination is to be represented as a single two-port network. The parameters are obtained by multiplying the individual

- (a) z-parameter matrix (c) y-parameter matrix
(b) h-parameter matrix (d) ABCD parameter matrix

CO3 1

11 Tellegen's theorem is applicable to:

- a) Non-linear, non-reciprocal networks only c) Any lumped network satisfying Kirchhoff's laws
b) Linear, time-invariant networks only d) Only DC circuits with independent sources

CO2 1

12 What does mutual inductance depend on?

- a) Resistance between the coils c) Voltage across the coils
b) Coupling coefficient and inductances d) Capacitance of the system

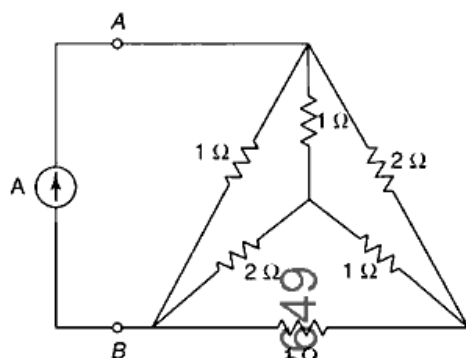
CO1 1

Q Solve the following.

- A) Classify circuit elements as lumped and distributed, linear and non-linear, and unilateral and bilateral. Provide suitable examples for each classification.
B) Find the voltage between terminals A and B in The given network.

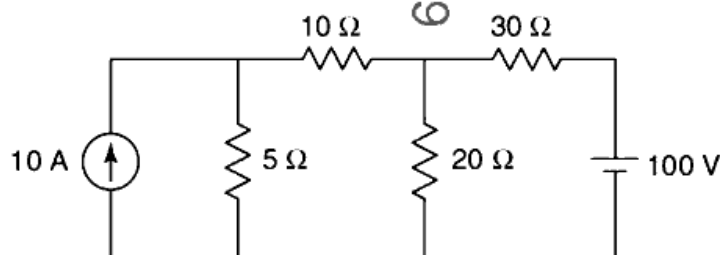
CO1 6

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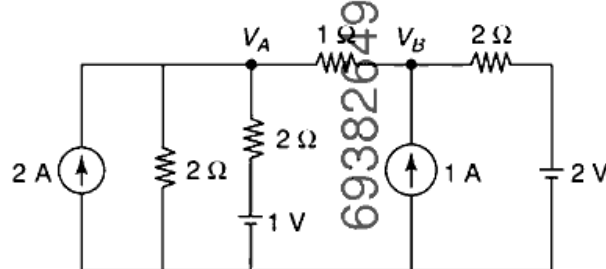


Q.3 Solve the following.

A) Find the current through the $10\ \Omega$ resistor in given network.

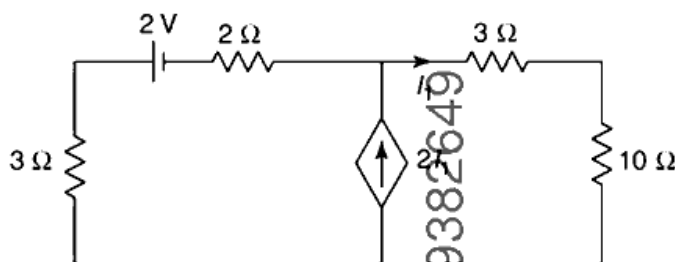


B) Find V_A and V_B in the network using node analysis.



Q.4 Solve Any Two of the following.

A) Find Norton's equivalent network and hence find the current in the $10\ \Omega$ resistor in Fig.



B) Define the terms graph, tree, branches, and chords in network topology

C) In the network of Fig., the switch is moved from the position 1 to 2 at $t = 0$, steady-state condition having been established in the position 1. Determine $i(t)$ for $t > 0$.

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CO2 6

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CO2 6

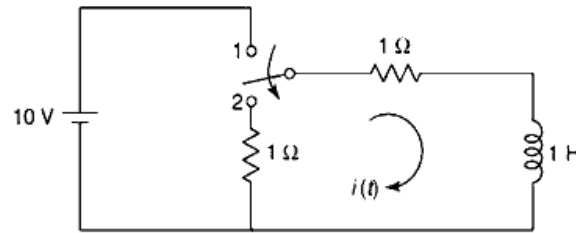
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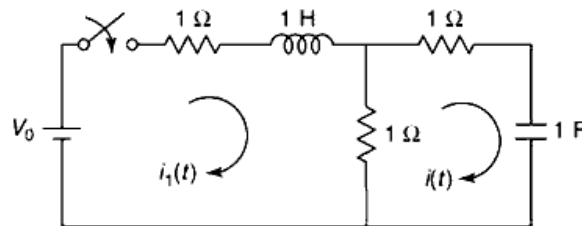
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CO4 6

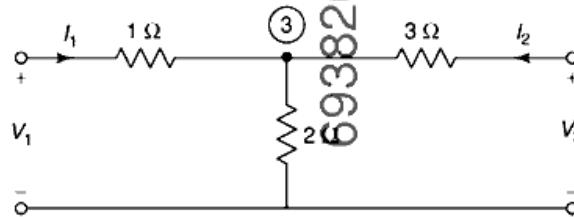


Q.5 Solve Any Two of the following.

- A) Obtain transient response of driven R-L series circuit.
- B) The switch K is closed at $t = 0$ and $V_0 = 1\text{ V}$ in the network shown in Fig. Determine $i(0^+)$, $\frac{di(0^+)}{dt}$, $\frac{di^2(0^+)}{dt^2}$



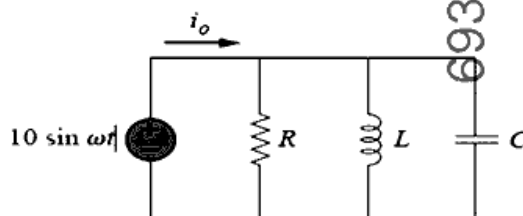
Find Y-parameters for the network shown



Q.6 Solve Any Two of the following.

- A) Explain low pass filter and high pass filter.
- B) Derive the expressions for Z-parameters and explain their significance in network analysis.

In the parallel RLC circuit, let $R = 8\text{ k}\Omega$, $L = 0.2\text{ mH}$, and $C = 8\text{ }\mu\text{F}$. (a) Calculate ω_0 , Q , and B . (b) Find ω_1 and ω_2 . (c) Determine the power dissipated at ω_0 , ω_1 , and ω_2 .



*** End ***